

Project Planning Phase Planning Logic

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	

Concept Overview

A Sprint is a fixed period during which a set of tasks is completed. An Epic is a large task broken down into smaller Stories, each with Story Points (Fibonacci sequence: 1, 2, 3, 5) representing effort.

Sprint 1 – Data Collection & Preprocessing

Epic	User Story	Task	Story Points
Epic 1: Data Collection	USN-1	Gather loan_data.csv (614 records)	2
Epic 2: Data Preprocessing	USN-2	Remove Loan_ID; handle missing values (mode/median)	3
Epic 2: Data Preprocessing	USN-3	Label-encode categorical variables	3
Epic 2: Data Preprocessing	USN-4	Scale numerical features with StandardScaler	2
Epic 2: Data Preprocessing	USN-5	80/20 train-test split	1

Total Story Points (Sprint 1): 11

Sprint 2 – Model Training & Comparison

Epic	User Story	Task	Story Points
Epic 3: Model Training	USN-6	Train Decision Tree, Random Forest, KNN, XGBoost	5
Epic 3: Model Training	USN-7	Evaluate with accuracy, classification report, confusion matrix	3
Epic 4: Model Selection	USN-8	Select best model (KNN – 84.55% accuracy)	2
Epic 4: Model Selection	USN-9	Save model.pkl, encoder.pkl, scaler.pkl	2

Total Story Points (Sprint 2): 12

Sprint 3 – Flask Application & UI

Epic	User Story	Task	Story Points
Epic 5: Web Application	USN-10	Build Home, Predict and About pages	3
Epic 5: Web Application	USN-11	Build Result page (eligibility %, confidence, risk, reasons)	5
Epic 5: Web Application	USN-12	Build Applicant Profile and History pages	3

Total Story Points (Sprint 3): 11

Velocity Calculation

Velocity = Total Story Points Completed ÷ Number of Sprints = 34 ÷ 3 ≈ 11.3

Team Velocity ≈ 11 Story Points per Sprint

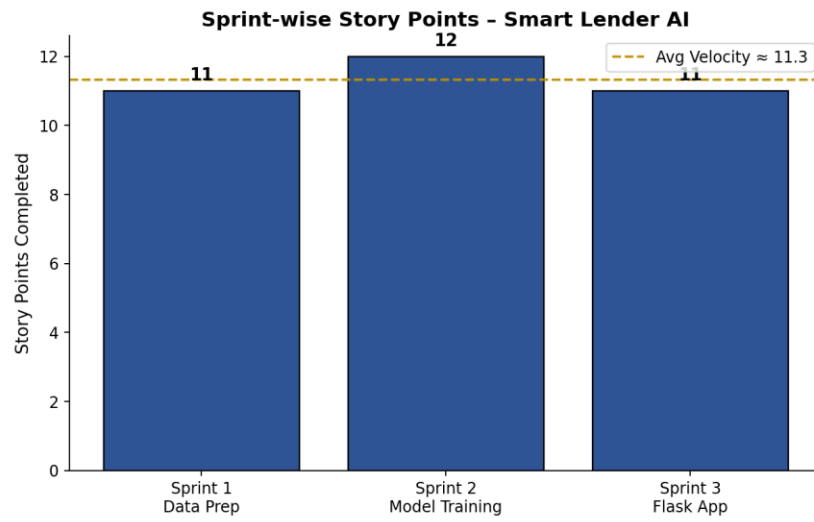


Fig 1: Sprint-wise Story Points Completed